

Tecniche Sperimentali per la Fisica delle Alte Energie

Configurazione Terminale e Istallazione Geant4

S. Germani – stefano.germani@unipg.it

Programma Lezioni

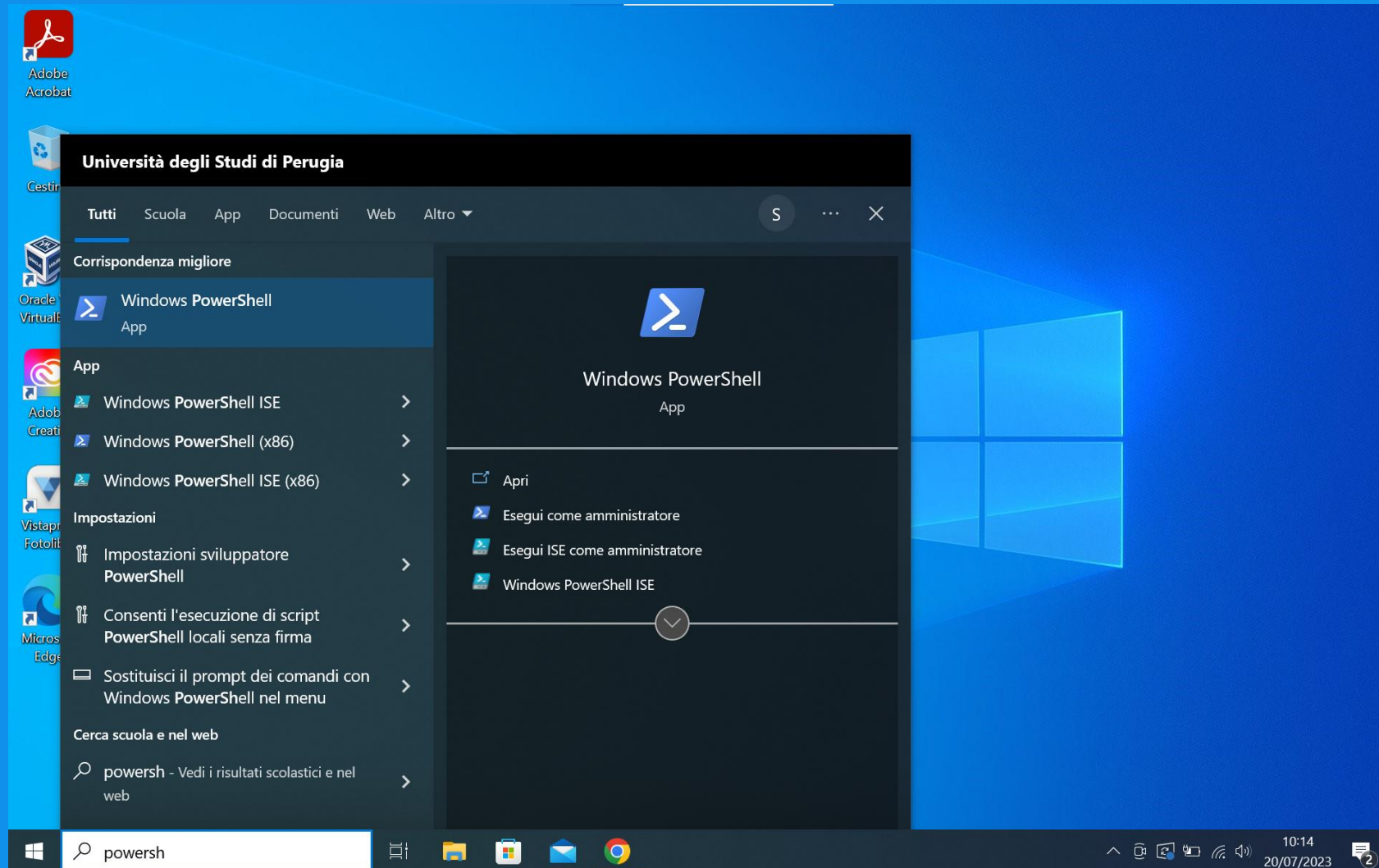
- **Lezione 1**

- Preparazione Ambiente di Lavoro
- Installazione Geant4 (+ Test Funzionamento)
- Introduzione a Geant 4

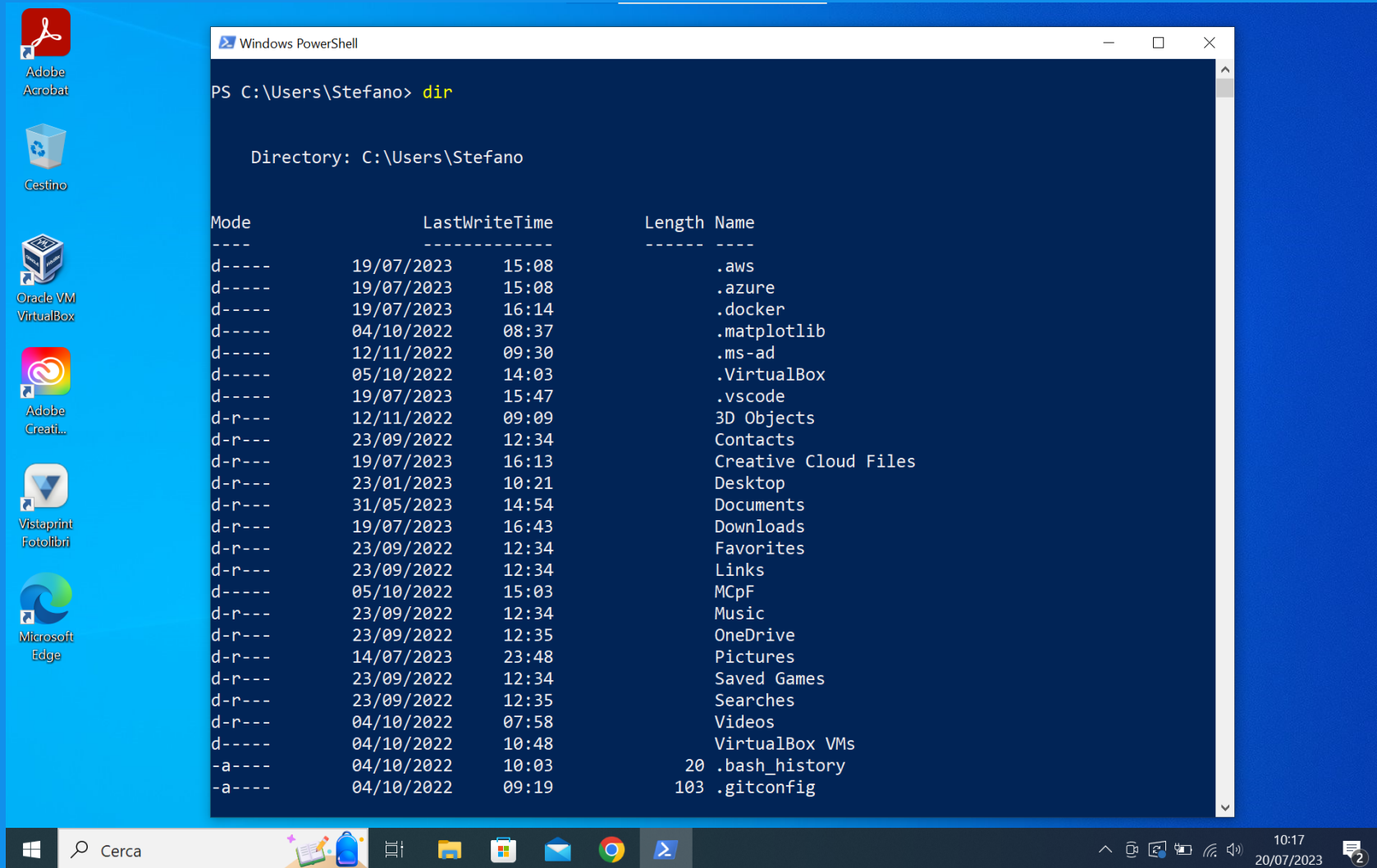
- **Lezione 2**

- Analisi Esempi
- Esercitazione

Windows PowerShell



Windows Power Shell

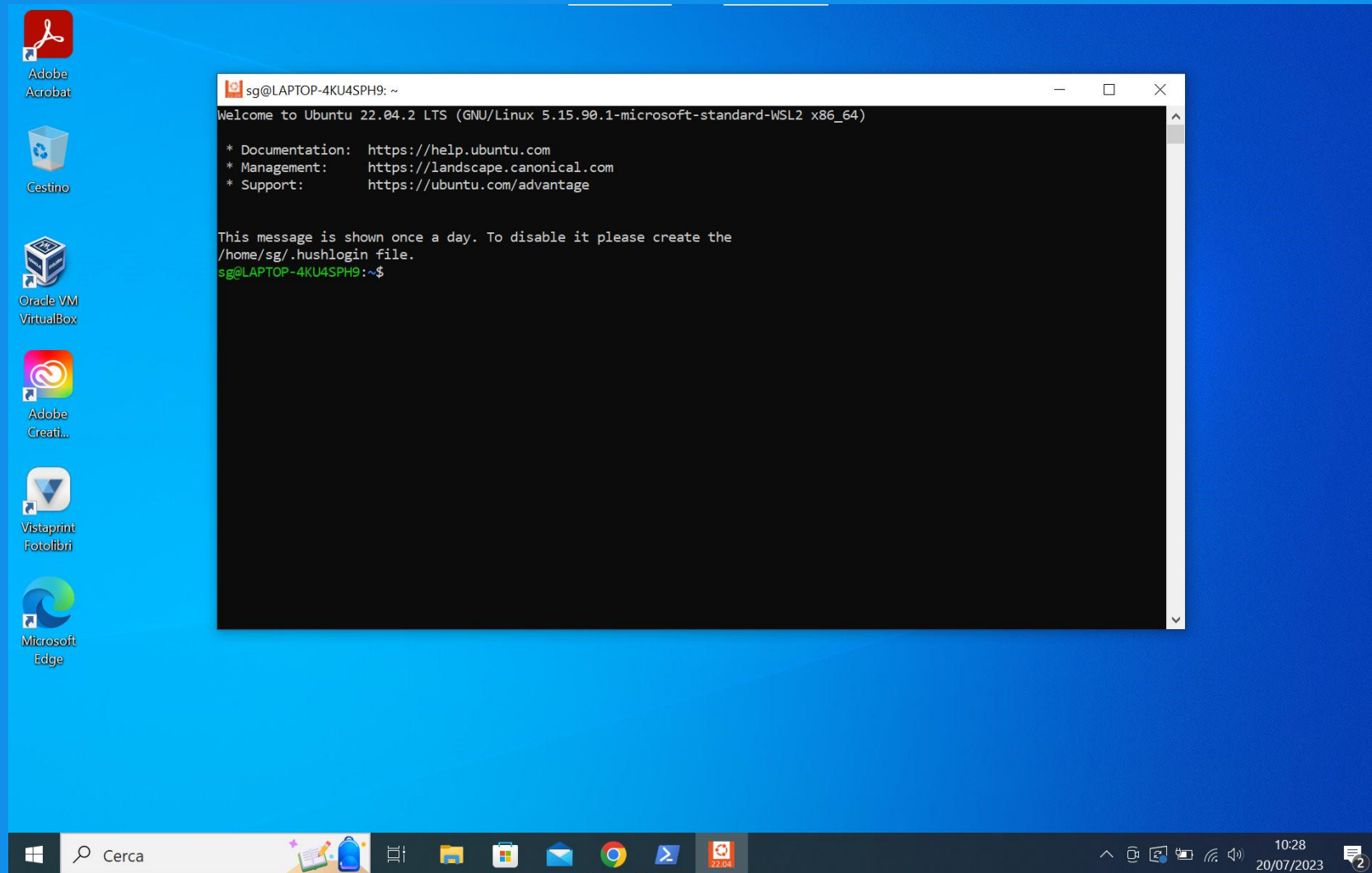


```
PS C:\Users\Stefano> dir

Directory: C:\Users\Stefano

Mode                LastWriteTime         Length Name
----                -
d-----          19/07/2023         15:08      .aws
d-----          19/07/2023         15:08      .azure
d-----          19/07/2023         16:14      .docker
d-----          04/10/2022          08:37      .matplotlib
d-----         12/11/2022          09:30      .ms-ad
d-----          05/10/2022         14:03      .VirtualBox
d-----          19/07/2023         15:47      .vscode
d-r---         12/11/2022          09:09      3D Objects
d-r---         23/09/2022         12:34      Contacts
d-r---          19/07/2023         16:13      Creative Cloud Files
d-r---         23/01/2023         10:21      Desktop
d-r---         31/05/2023         14:54      Documents
d-r---          19/07/2023         16:43      Downloads
d-r---         23/09/2022         12:34      Favorites
d-r---         23/09/2022         12:34      Links
d-----          05/10/2022         15:03      MCpF
d-r---         23/09/2022         12:34      Music
d-r---         23/09/2022         12:35      OneDrive
d-r---         14/07/2023         23:48      Pictures
d-r---         23/09/2022         12:34      Saved Games
d-r---         23/09/2022         12:35      Searches
d-r---          04/10/2022          07:58      Videos
d-----          04/10/2022         10:48      VirtualBox VMs
-a---          04/10/2022          10:03      20 .bash_history
-a---          04/10/2022          09:19     103 .gitconfig
```

Windows Subsystem for Linux (WSL)



WSL

<https://learn.microsoft.com/it-it/windows/wsl/install>

The screenshot shows a web browser window displaying the Microsoft Learn page for installing Linux on Windows using WSL. The page is in Italian and features a navigation bar with the Microsoft logo, 'Learn', and various links like 'Documentazione', 'Formazione', 'Certificazioni', 'Q&A', 'Esempi di codice', and 'Più informazioni'. A search bar and a 'Accedi' button are also present. The main content area is titled 'Installare Linux in Windows con WSL' and includes a breadcrumb trail: 'Learn / Windows / Ambiente di sviluppo / WSL /'. Below the title, it says 'Articolo • 07/07/2023 • 8 contributori' and has a 'Commenti e suggerimenti' button. A section titled 'In questo articolo' lists links for 'Prerequisiti', 'Installare il comando WSL', 'Modificare la distribuzione Linux predefinita installata', and 'Configurare le informazioni utente Linux', followed by a 'Mostra 6 in più' link. The main text begins with 'Gli sviluppatori possono accedere alla potenza di Windows e Linux contemporaneamente in un computer Windows. La sottosistema Windows per Linux (WSL) consente agli sviluppatori di installare una distribuzione Linux (ad esempio Ubuntu, OpenSUSE, Kali, Debian, Arch Linux e così via) e usare'. On the left, a sidebar shows a filter 'Filtra in base al titolo' and a list of topics under 'Documentazione di WSL', with 'Installare WSL' selected. On the right, a 'Risorse aggiuntive' section lists 'Documentazione', 'Configurare un ambiente di sviluppo WSL', 'Confronto tra versioni di WSL', and 'Risoluzione dei problemi relativi a Sottosistema Windows per Linux', each with a brief description and a 'Mostra X in più' link.

Installare WSL | Microsoft Learn x +

learn.microsoft.com/it-it/windows/wsl/install

Microsoft | Learn Documentazione Formazione Certificazioni Q&A Esempi di codice Più informazioni v

Cerca Accedi

Windows Integrità rilascio Client Windows v Sviluppatori di applicazioni v Sviluppatori di hardware v Windows Server Windows per IoT Più informazioni v

Filtra in base al titolo

Documentazione di WSL

- > Panoramica
- > Installazione
 - Installare WSL**
 - Passaggi di installazione manuali per le versioni precedenti
 - Installare in Windows Server
- > Esercitazioni
- > Concetti
- > Procedure
- Domande frequenti
- Risoluzione dei problemi
- > Note sulla versione

Scarica il PDF

Learn / Windows / Ambiente di sviluppo / WSL /

Installare Linux in Windows con WSL

Articolo • 07/07/2023 • 8 contributori

Commenti e suggerimenti

In questo articolo

- Prerequisiti
- Installare il comando WSL
- Modificare la distribuzione Linux predefinita installata
- Configurare le informazioni utente Linux

Mostra 6 in più

Gli sviluppatori possono accedere alla potenza di Windows e Linux contemporaneamente in un computer Windows. La sottosistema Windows per Linux (WSL) consente agli sviluppatori di installare una distribuzione Linux (ad esempio Ubuntu, OpenSUSE, Kali, Debian, Arch Linux e così via) e usare

Risorse aggiuntive

Documentazione

Configurare un ambiente di sviluppo WSL

Configurare un ambiente di sviluppo WSL usando le procedure consigliate di questa guida dettagliata. Informazioni su come...

Confronto tra versioni di WSL

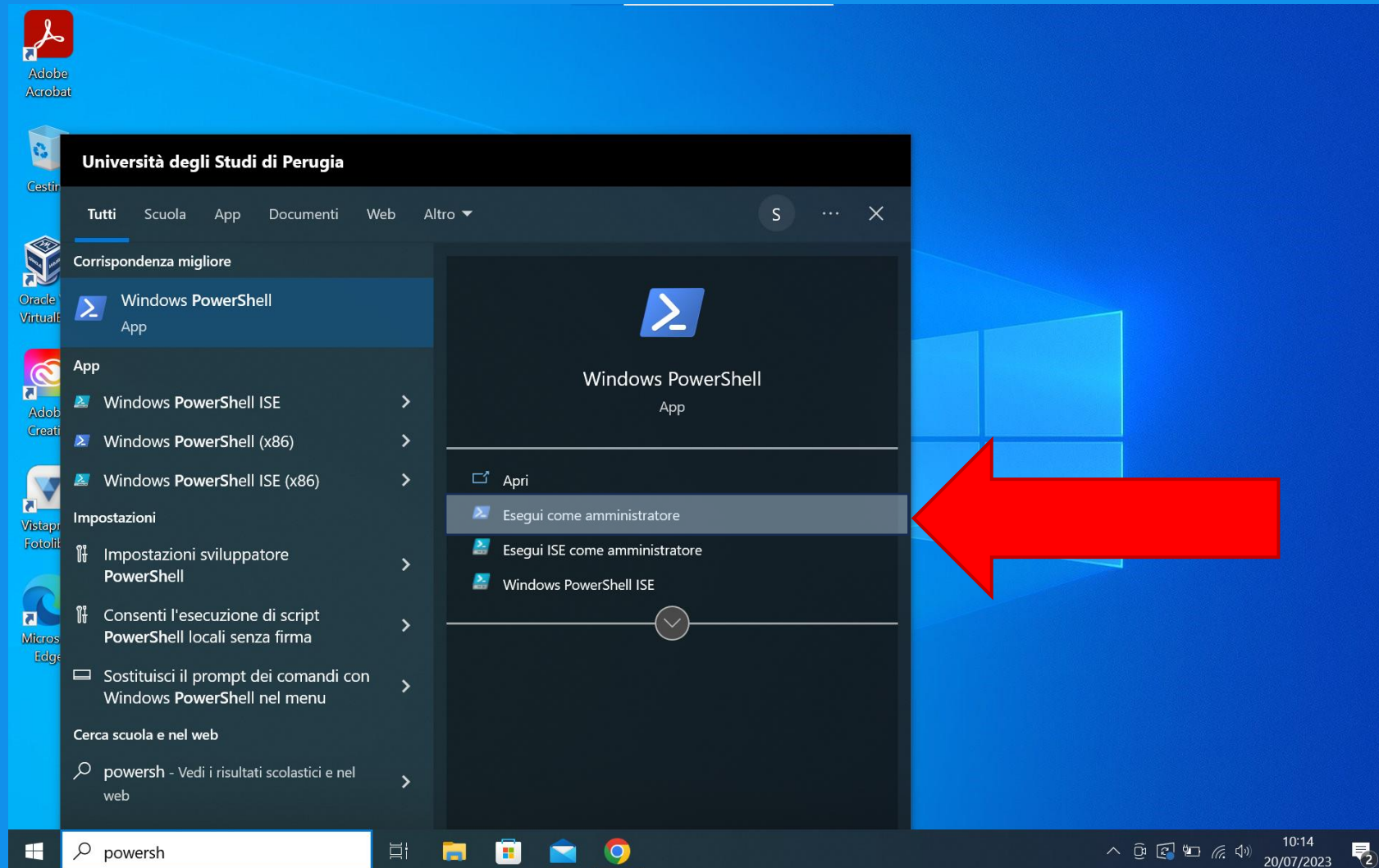
WSL 2 offre i vantaggi di WSL 1, ma usa un kernel Linux effettivo, anziché un livello di conversione come WSL 1, con...

Risoluzione dei problemi relativi a Sottosistema Windows per Linux

Fornisce informazioni dettagliate sugli errori e sui problemi comuni che si verificano durante l'esecuzione di Linux i...

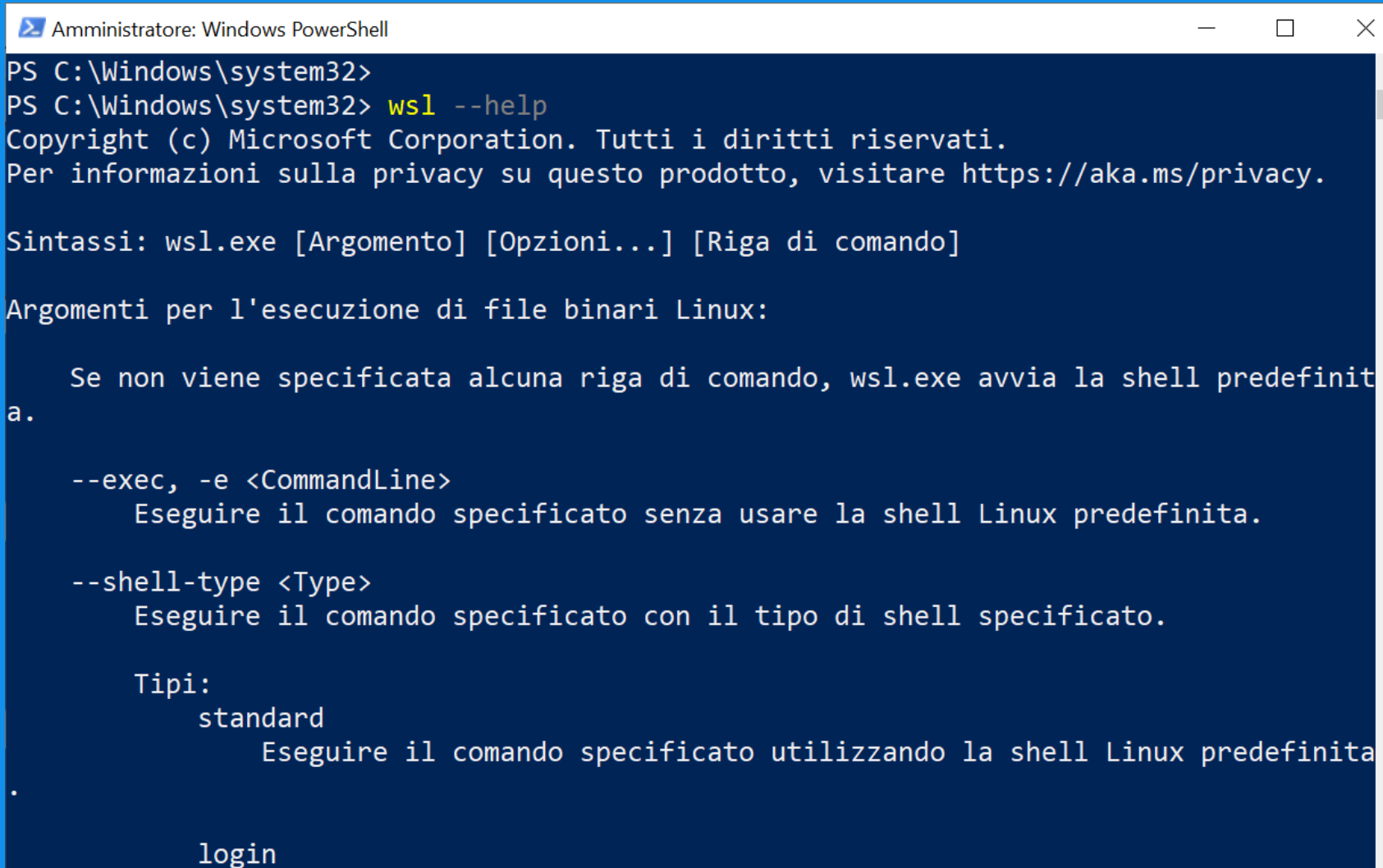
Mostra 5 in più

Windows PowerShell come Amministratore



Se richiedo concedere
il permesso di
apportare modifiche
al dispositivo

WSL

A screenshot of a Windows PowerShell terminal window titled "Amministratore: Windows PowerShell". The terminal shows the command `wsl --help` being executed. The output displays copyright information for Microsoft Corporation, a link to the privacy policy, and the syntax for the `wsl.exe` command. It also lists arguments for Linux binary execution, including `--exec` and `--shell-type`, and lists shell types: `standard` and `login`.

```
Amministratore: Windows PowerShell
PS C:\Windows\system32>
PS C:\Windows\system32> wsl --help
Copyright (c) Microsoft Corporation. Tutti i diritti riservati.
Per informazioni sulla privacy su questo prodotto, visitare https://aka.ms/privacy.

Sintassi: wsl.exe [Argomento] [Opzioni...] [Riga di comando]

Argomenti per l'esecuzione di file binari Linux:

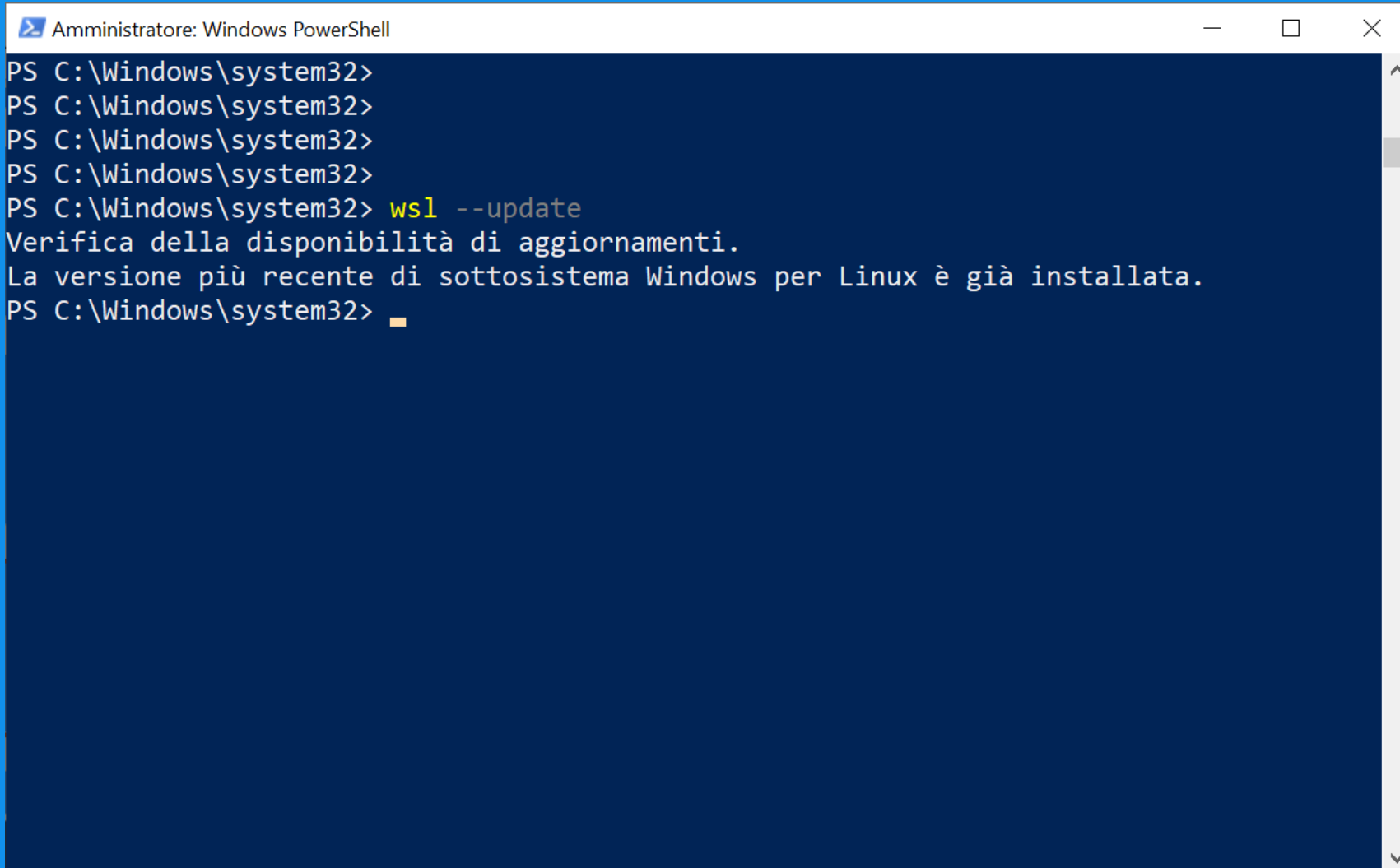
    Se non viene specificata alcuna riga di comando, wsl.exe avvia la shell predefinita.

    --exec, -e <CommandLine>
        Eseguire il comando specificato senza usare la shell Linux predefinita.

    --shell-type <Type>
        Eseguire il comando specificato con il tipo di shell specificato.

    Tipi:
        standard
            Eseguire il comando specificato utilizzando la shell Linux predefinita.
        login
```

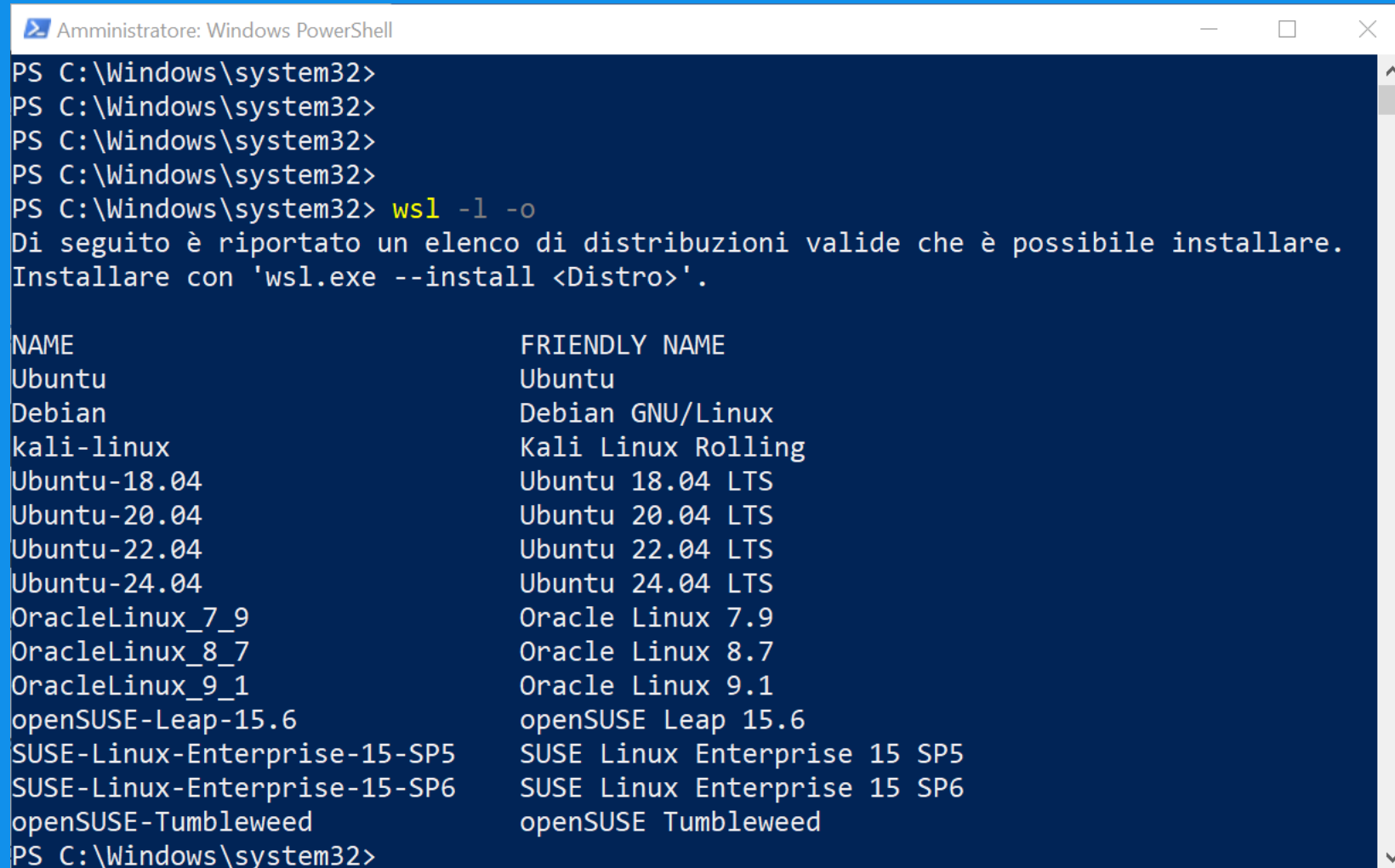

WSL



```
Amministratore: Windows PowerShell
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> wsl --update
Verifica della disponibilità di aggiornamenti.
La versione più recente di sottosistema Windows per Linux è già installata.
PS C:\Windows\system32> █
```

WSL

Aprire PowerShell come Amministratore

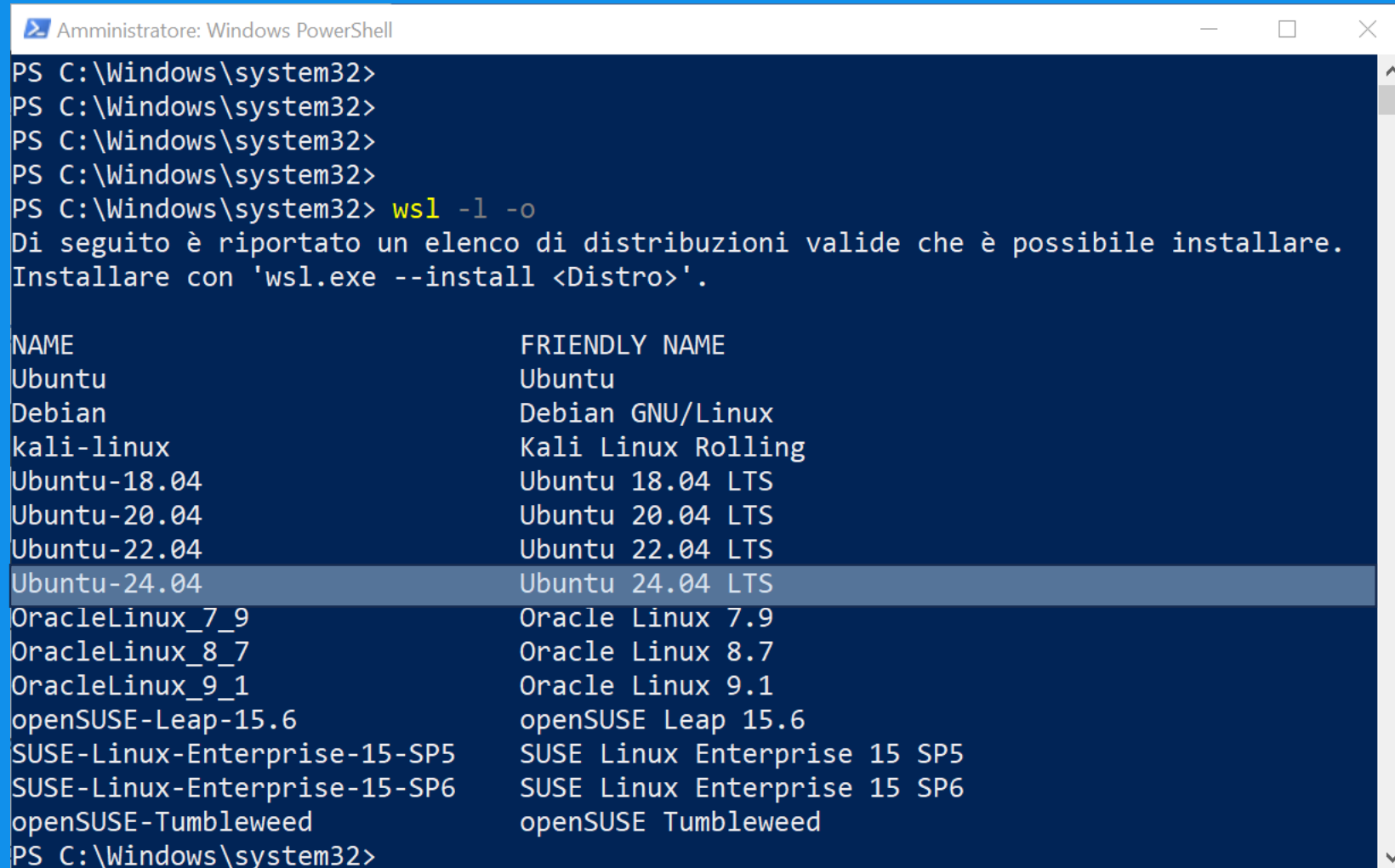


```
Amministratore: Windows PowerShell
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> wsl -l -o
Di seguito è riportato un elenco di distribuzioni valide che è possibile installare.
Installare con 'wsl.exe --install <Distro>'.

NAME                                FRIENDLY NAME
-----                                -
Ubuntu                              Ubuntu
Debian                              Debian GNU/Linux
kali-linux                          Kali Linux Rolling
Ubuntu-18.04                        Ubuntu 18.04 LTS
Ubuntu-20.04                        Ubuntu 20.04 LTS
Ubuntu-22.04                        Ubuntu 22.04 LTS
Ubuntu-24.04                        Ubuntu 24.04 LTS
OracleLinux_7_9                     Oracle Linux 7.9
OracleLinux_8_7                     Oracle Linux 8.7
OracleLinux_9_1                     Oracle Linux 9.1
openSUSE-Leap-15.6                  openSUSE Leap 15.6
SUSE-Linux-Enterprise-15-SP5        SUSE Linux Enterprise 15 SP5
SUSE-Linux-Enterprise-15-SP6        SUSE Linux Enterprise 15 SP6
openSUSE-Tumbleweed                 openSUSE Tumbleweed
PS C:\Windows\system32>
```

WSL

Aprire PowerShell come Amministratore



```
Amministratore: Windows PowerShell
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> wsl -l -o
Di seguito è riportato un elenco di distribuzioni valide che è possibile installare.
Installare con 'wsl.exe --install <Distro>'.

NAME                                FRIENDLY NAME
-----                                -
Ubuntu                              Ubuntu
Debian                              Debian GNU/Linux
kali-linux                          Kali Linux Rolling
Ubuntu-18.04                         Ubuntu 18.04 LTS
Ubuntu-20.04                         Ubuntu 20.04 LTS
Ubuntu-22.04                         Ubuntu 22.04 LTS
Ubuntu-24.04                         Ubuntu 24.04 LTS
OracleLinux_7_9                     Oracle Linux 7.9
OracleLinux_8_7                     Oracle Linux 8.7
OracleLinux_9_1                     Oracle Linux 9.1
openSUSE-Leap-15.6                  openSUSE Leap 15.6
SUSE-Linux-Enterprise-15-SP5        SUSE Linux Enterprise 15 SP5
SUSE-Linux-Enterprise-15-SP6        SUSE Linux Enterprise 15 SP6
openSUSE-Tumbleweed                 openSUSE Tumbleweed
PS C:\Windows\system32>
```

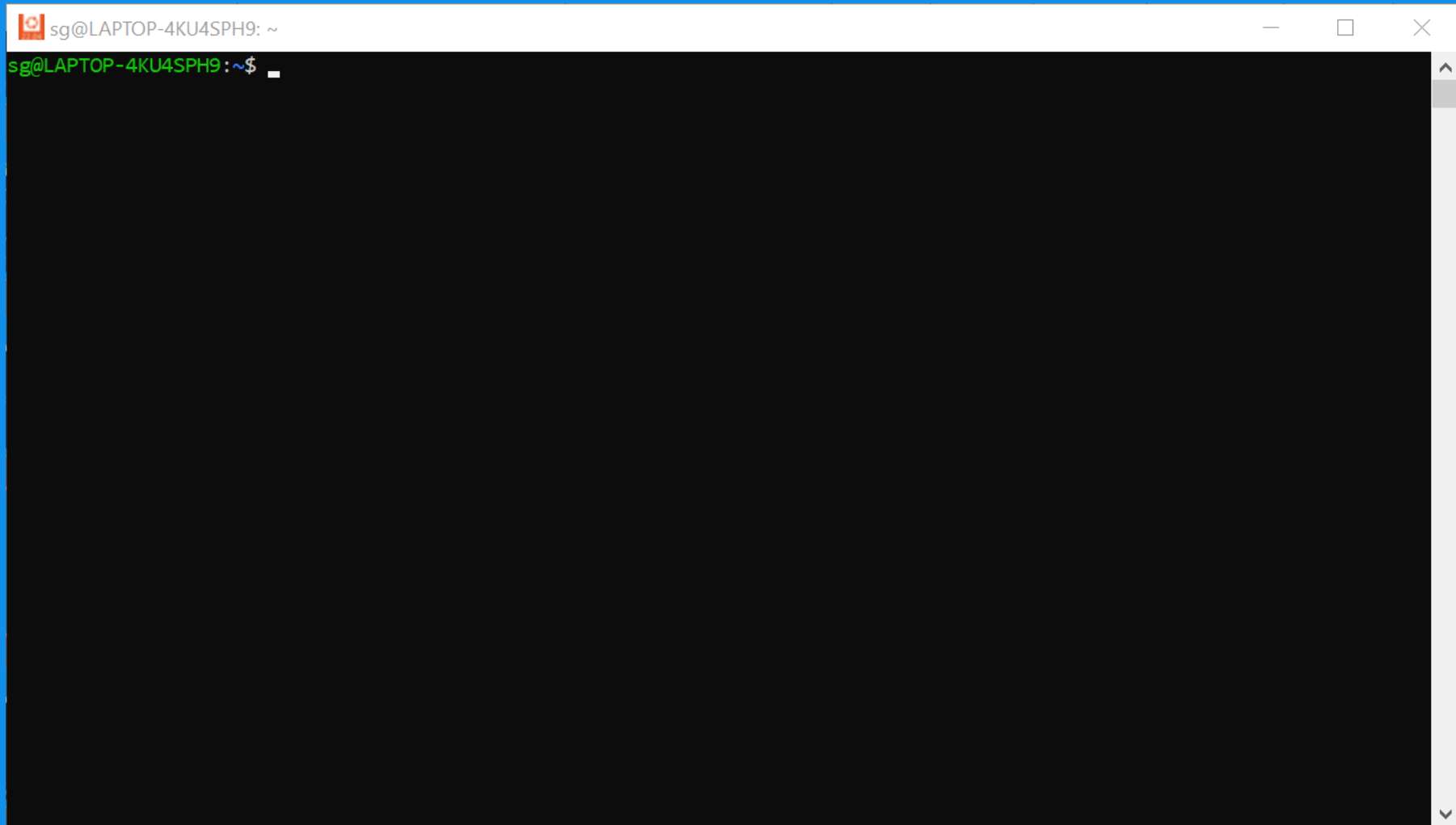
WSL

```
Amministratore: Windows PowerShell

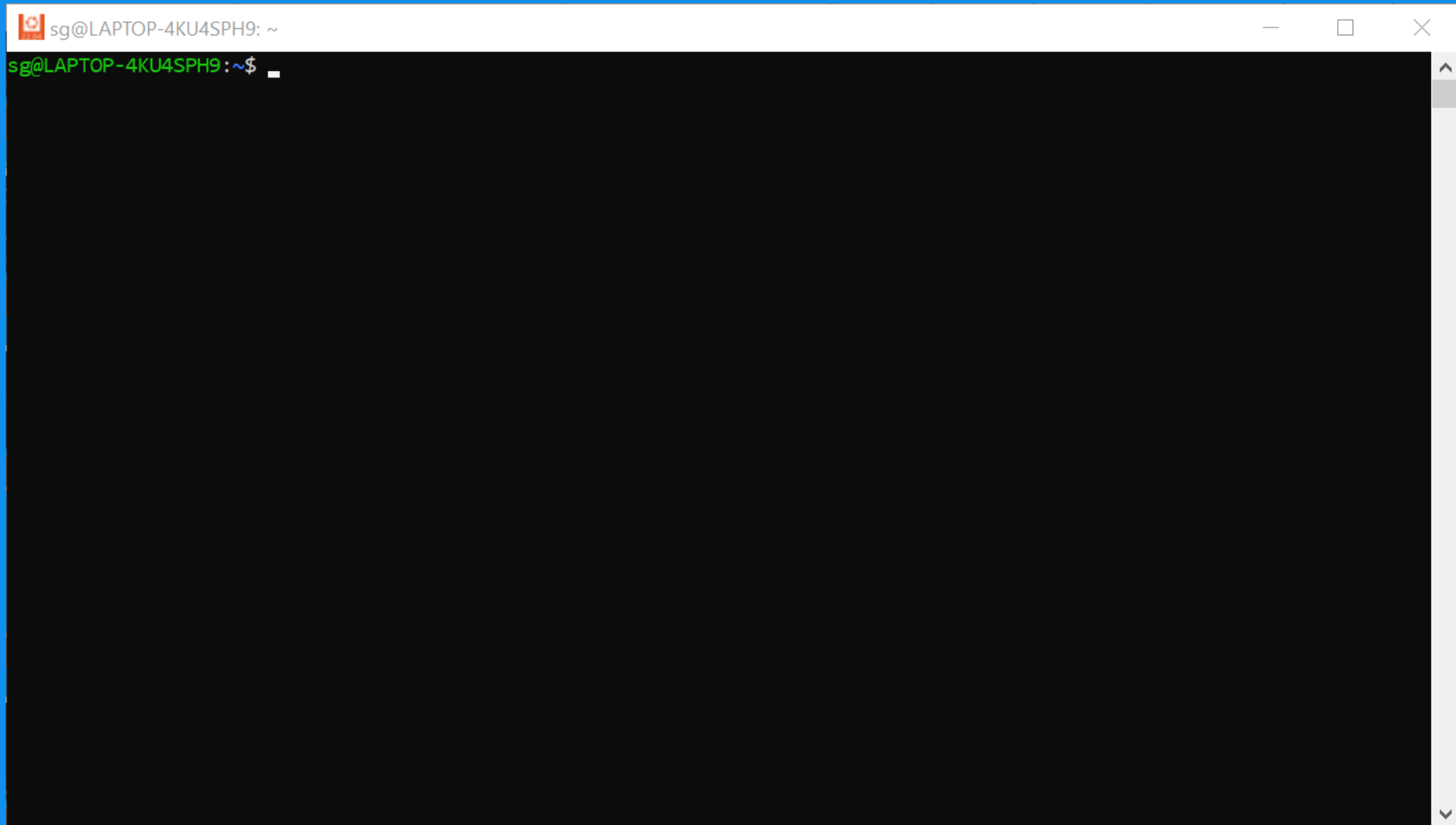
PS C:\Windows\system32> wsl -l -o
Di seguito è riportato un elenco di distribuzioni valide che è possibile installare.
Installare con 'wsl.exe --install <Distro>'.

NAME                                FRIENDLY NAME
Ubuntu                             Ubuntu
Debian                             Debian GNU/Linux
kali-linux                         Kali Linux Rolling
Ubuntu-18.04                       Ubuntu 18.04 LTS
Ubuntu-20.04                       Ubuntu 20.04 LTS
Ubuntu-22.04                       Ubuntu 22.04 LTS
Ubuntu-24.04                       Ubuntu 24.04 LTS
OracleLinux_7_9                   Oracle Linux 7.9
OracleLinux_8_7                   Oracle Linux 8.7
OracleLinux_9_1                   Oracle Linux 9.1
openSUSE-Leap-15.6                openSUSE Leap 15.6
SUSE-Linux-Enterprise-15-SP5      SUSE Linux Enterprise 15 SP5
SUSE-Linux-Enterprise-15-SP6      SUSE Linux Enterprise 15 SP6
openSUSE-Tumbleweed               openSUSE Tumbleweed
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> wsl --install -d Ubuntu-24.04
```

Terminale Linux WLS



UPDATE ...



Prerequisiti

[illegible]

Prerequisiti

[illegible]

Download

<https://geant4.web.cern.ch/>

GEANT4
A SIMULATION TOOLKIT

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Geant4

Toolkit for the simulation of the passage of particles through matter. Its areas of application include high energy, nuclear and accelerator physics, as well as studies in medical and space science.



Get started

Everything you need to get started with Geant4.

[I'm ready to start!](#)



Download

Geant4 source code and installers are available for download, with source code under an [open source license](#).

Latest: [11.2.2](#)



Docs

Documentation for Geant4, along with tutorials and guides, are available online.

[Read documentation](#)

Download

<https://geant4.web.cern.ch/download/11.3.2.html>

Download Geant4-11.3.2

First released 28 Apr 2025

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RELEASE NOTES

See:

[Main Release Notes](#) - [Patch-1](#) - [Patch-2](#) -

Source code

Source code is freely available from [CERN GitLab](#) or through [GitHub](#).

Source code can also be browsed through the [LXR source code browser](#).

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Download tar.gz

Download tar.bz2

Download tar

Download

<https://geant4.web.cern.ch/download/11.3.2.html>

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RELEASE NOTES

See:

[Main Release Notes](#) - [Patch-1](#) - [Patch-2](#)

Source code

Source code is freely available from

Source code can also be browsed th

Apri link in un'altra scheda

Apri link in un'altra finestra

Apri link in finestra di navigazione in incognito

Salva link con nome...

Copia indirizzo link

Ispeziona

Download zip

Download tar.gz

Download tar.bz2

Download tar

Binary releases

<https://gitlab.cern.ch/geant4/geant4/-/archive/v11.3.2/geant4-v11.3.2.tar.gz>

Prerequisiti

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
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sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$ wget https://gitlab.cern.ch/geant4/geant4/-/archive/v11.3.2/geant4-v11.3.2.tar.gz
```

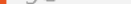
Spacchettamento

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4
sg@LAPTOP-4KU4SPH9: ~/GEANT4$
sg@LAPTOP-4KU4SPH9: ~/GEANT4$
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sg@LAPTOP-4KU4SPH9: ~/GEANT4$
sg@LAPTOP-4KU4SPH9: ~/GEANT4$
sg@LAPTOP-4KU4SPH9: ~/GEANT4$
sg@LAPTOP-4KU4SPH9: ~/GEANT4$ ls
geant4-v11.3.2.tar.gz
sg@LAPTOP-4KU4SPH9: ~/GEANT4$
sg@LAPTOP-4KU4SPH9: ~/GEANT4$
sg@LAPTOP-4KU4SPH9: ~/GEANT4$ tar -xzf geant4-v11.3.2.tar.gz
```

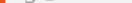
Ispezione Carella Codice Sorgente

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$ ls
geant4-v11.3.2.tar.gz
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$ tar -xzf geant4-v11.3.2.tar.gz
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$ ls -ltr
total 35908
drwxr-xr-x 11 sg sg      4096 Apr 28 13:49 geant4-v11.3.2
-rw-r--r--  1 sg sg 36761780 Oct 27 12:20 geant4-v11.3.2.tar.gz
sg@LAPTOP-4KU4SPH9:~/GEANT4$ ls -ltr geant4-v11.3.2
total 52
drwxr-xr-x 24 sg sg 4096 Apr 28 13:49 source
drwxr-xr-x  3 sg sg 4096 Apr 28 13:49 packaging
drwxr-xr-x  5 sg sg 4096 Apr 28 13:49 examples
drwxr-xr-x  2 sg sg 4096 Apr 28 13:49 environments
drwxr-xr-x  3 sg sg 4096 Apr 28 13:49 config
drwxr-xr-x  4 sg sg 4096 Apr 28 13:49 cmake
drwxr-xr-x  2 sg sg 4096 Apr 28 13:49 ReleaseNotes
-rw-r--r--  1 sg sg 1984 Apr 28 13:49 README.rst
-rw-r--r--  1 sg sg 4029 Apr 28 13:49 LICENSE
-rw-r--r--  1 sg sg  990 Apr 28 13:49 CONTRIBUTING.rst
-rw-r--r--  1 sg sg 2193 Apr 28 13:49 CMakeLists.txt
-rw-r--r--  1 sg sg 5850 Apr 28 13:49 CITATION.cff
lrwxrwxrwx  1 sg sg   12 Apr 28 13:49 CHANGELOG -> ReleaseNotes
sg@LAPTOP-4KU4SPH9:~/GEANT4$
```


Cartelle per Istallazione



```
sg@LAPTOP-4KU4SPH9: ~/GEANT4
```

[illegible]

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4
```

[illegible]

Cartelle per Istallazione

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4
sg@LAPTOP-4KU4SPH9:~/GEANT4$
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sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$
sg@LAPTOP-4KU4SPH9:~/GEANT4$ ls
geant4-v11.3.2 geant4-v11.3.2-build geant4-v11.3.2-install geant4-v11.3.2.tar.gz
sg@LAPTOP-4KU4SPH9:~/GEANT4$
```

Istallazione

```
sg@LAPTOP-4KU4SPH9: ~/Gi × + ▾
-- Building PTL with option BUILD_SHARED_LIBS - ON
-- Building PTL with option BUILD_OBJECT_LIBS -
-- Building PTL with CMAKE_CXX_STANDARD - 17
-- Building PTL with option PTL_USE_COVERAGE -
-- Building PTL with option PTL_USE_SANITIZER -
-- Building PTL with option PTL_USE_CLANG_TIDY -
-- Building PTL with option PTL_USE_TBB - OFF
-- Building PTL with option PTL_USE_LOCKS - OFF
-- Building PTL with option PTL_INSTALL_HEADERS - ON
-- Building PTL with option PTL_INSTALL_CONFIG - ON
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- The following Geant4 features are enabled:
CMAKE_CXX_STANDARD: Compiling against C++ Standard '17'
GEANT4_BUILD_MULTITHREADED: Build multithread enabled libraries
GEANT4_BUILD_TLS_MODEL: Building with TLS model 'initial-exec'
GEANT4_USE_SYSTEM_EXPAT: Using system EXPAT library

-- Configuring done (3.3s)
-- Generating done (0.4s)
-- Build files have been written to: /home/sg/GEANT4/geant4-v11.3.2-build
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$ cmake -DGEANT4_INSTALL_DATA=ON .
```

Istallazione

```
sg@LAPTOP-4KU4SPH9: ~/Gi × + v
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$ cmake -DGEANT4_INSTALL_DATA=ON .
-- Checking C++ feature CXXSTDLIB_FILESYSTEM_NATIVE - Success
-- Configuring download of missing dataset G4NDL (4.7.1)
-- Configuring download of missing dataset G4EMLOW (8.6.1)
-- Configuring download of missing dataset PhotonEvaporation (6.1)
-- Configuring download of missing dataset RadioactiveDecay (6.1.2)
-- Configuring download of missing dataset G4PARTICLEXS (4.1)
-- Configuring download of missing dataset G4PII (1.3)
-- Configuring download of missing dataset RealSurface (2.2)
-- Configuring download of missing dataset G4SAIDDATA (2.0)
-- Configuring download of missing dataset G4ABLA (3.3)
-- Configuring download of missing dataset G4INCL (1.2)
-- Configuring download of missing dataset G4ENSDFSTATE (3.0)
-- Configuring download of missing dataset G4CHANNELING (1.0)
-- The following Geant4 features are enabled:
CMAKE_CXX_STANDARD: Compiling against C++ Standard '17'
GEANT4_BUILD_MULTITHREADED: Build multithread enabled libraries
GEANT4_BUILD_TLS_MODEL: Building with TLS model 'initial-exec'
GEANT4_USE_SYSTEM_EXPAT: Using system EXPAT library

-- Configuring done (1.8s)
-- Generating done (0.3s)
-- Build files have been written to: /home/sg/GEANT4/geant4-v11.3.2-build
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$ make -j 4
```

Istallazione

```
sg@LAPTOP-4KU4SPH9: ~/GI  ×  +  ▾

[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/FTFP_BERT_TRV.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/FTFQGSP_BERT.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/G4PhysListFactoryAlt.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/G4PhysListFactory.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/G4PhysListFactoryMessenger.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/G4PhysListRegistry.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/LBE.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/NuBeam.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QBBC.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QBBC_ABLA.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGS_BIC.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_BERT.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_BERT_HP.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_BIC_AllHP.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_BIC.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_BIC_HP.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_BIC_HPT.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/QGSP_FTFP_BERT.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/lists/src/Shielding.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/util/src/G4HadParticles.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/util/src/G4HadProcesses.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/util/src/G4PhysListUtil.cc.o
[100%] Building CXX object source/CMakeFiles/G4physicslists.dir/physics_lists/util/src/G4WarnPLStatus.cc.o
[100%] Linking CXX shared library ../BuildProducts/lib/libG4physicslists.so
[100%] Built target G4physicslists
[100%] Linking CXX shared library ../BuildProducts/lib/libG4ToolsSG.so
[100%] Built target G4ToolsSG
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$ |
```

Installazione

[illegible]

Installazione

```
sg@LAPTOP-4KU4SPH9: ~/GI  ×  +  ▾  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02EventAction.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02TrackingAction.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02PhysicsList.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02FastSimModelHCal.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02FastSimModelEMCal.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02RunAction.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02DetectorParametrisation.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02PrimaryParticleInformation.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02Output.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02DetectorConstruction.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/include/Par  
02EventInformation.hh  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/examplePar0  
2.out  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/examplePar0  
2.cc  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/Par02/.README.txt  
-- Installing: /home/sg/GEANT4/geant4-v11.3.2-install/share/Geant4/examples/extended/parameterisations/.README.txt  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2-build$
```

Installazione

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$  
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2-build$ cd  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ ls  
GEANT4  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ source GEANT4/geant4-v11.3.2-install/bin/geant4.sh |
```


Prova Esecuzione Esempio

Documentazione Esempi:

https://geant4-userdoc.web.cern.ch/Doxygen/examples_doc/html/index.html

How to build and run an example

Compile and link to generate an executable

```
% cd path_to_exampleXYZ      # go to directory which contains your example
% mkdir exampleXYZ_build
% cd exampleXYZ_build
% cmake -DGeant4_DIR=path_to_Geant4_installation/lib[64]/cmake/Geant4/ ../exampleXYZ
% make -j N exampleXYZ       # "N" is the number of processes
% make install               # this step is optional
```

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gt × + ∨  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$ mkdir B1_build|
```

Prova Esempio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gi × + v
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$ cd B1_build/
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ pwd
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ls /home/sg/GEANT4/geant4-v11.3.2-install/lib/cmake/Geant4/
G4EXPATShim.cmake      G4MotifShim.cmake      Geant4ConfigVersion.cmake      Modules
G4FreetypeShim.cmake   G4X11Shim.cmake        Geant4LibraryDepends-release.cmake PTL
G4HDF5Shim.cmake       Geant4Config.cmake     Geant4LibraryDepends.cmake     UseGeant4.cmake
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ cmake -DGeant4_DIR=/home/sg/GEANT4/geant4-v11.3.2
-install/lib/cmake/Geant4/ ../|
```

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ make -j 4 exampleB1  
[ 12%] Building CXX object CMakeFiles/exampleB1.dir/exampleB1.cc.o  
[ 37%] Building CXX object CMakeFiles/exampleB1.dir/src/DetectorConstruction.cc.o  
[ 25%] Building CXX object CMakeFiles/exampleB1.dir/src/ActionInitialization.cc.o  
[ 50%] Building CXX object CMakeFiles/exampleB1.dir/src/EventAction.cc.o  
[ 62%] Building CXX object CMakeFiles/exampleB1.dir/src/PrimaryGeneratorAction.cc.o  
[ 75%] Building CXX object CMakeFiles/exampleB1.dir/src/RunAction.cc.o  
[ 87%] Building CXX object CMakeFiles/exampleB1.dir/src/SteppingAction.cc.o  
[100%] Linking CXX executable exampleB1  
[100%] Built target exampleB1  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ |
```

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gi × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ls  
CMakeCache.txt  Makefile          exampleB1      exampleB1.out  run1.mac    vis.mac  
CMakeFiles     cmake_install.cmake  exampleB1.in  init_vis.mac   run2.mac  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ |
```


Prova Esemplio B1

[illegible]

```
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ./exampleB1 run1.mac
```

```
*****  
Geant4 version Name: geant4-11-03-patch-02 [MT] (25-April-2025)  
<< in Multi-threaded mode >>  
      Copyright : Geant4 Collaboration  
      References : NIM A 506 (2003), 250-303  
                  : IEEE-TNS 53 (2006), 270-278  
                  : NIM A 835 (2016), 186-225  
      WWW : http://geant4.org/  
*****  
  
<<< Reference Physics List QBBC  
Accumulable registered as "accumulable_0"  
Accumulable registered as "accumulable_1"  
Visualization Manager instantiating with verbosity "warnings (3)"...  
Visualization Manager initialising...  
Registering graphics systems...  
  
You have successfully registered the following graphics systems.  
Registered graphics systems are:  
  ASCIIITree (ATree)  
  DAWNFILE (DAWNFILE)  
  G4HepRepFile (HepRepFile)  
  RayTracer (RayTracer)  
  VRML2FILE (VRML2FILE)  
  gMocrenFile (gMocrenFile)  
  TOOLSSG_OFFSCREEN (TSG_OFFSCREEN, TSG_FILE, TSG)  
You may choose a graphics system (driver) with a parameter of
```

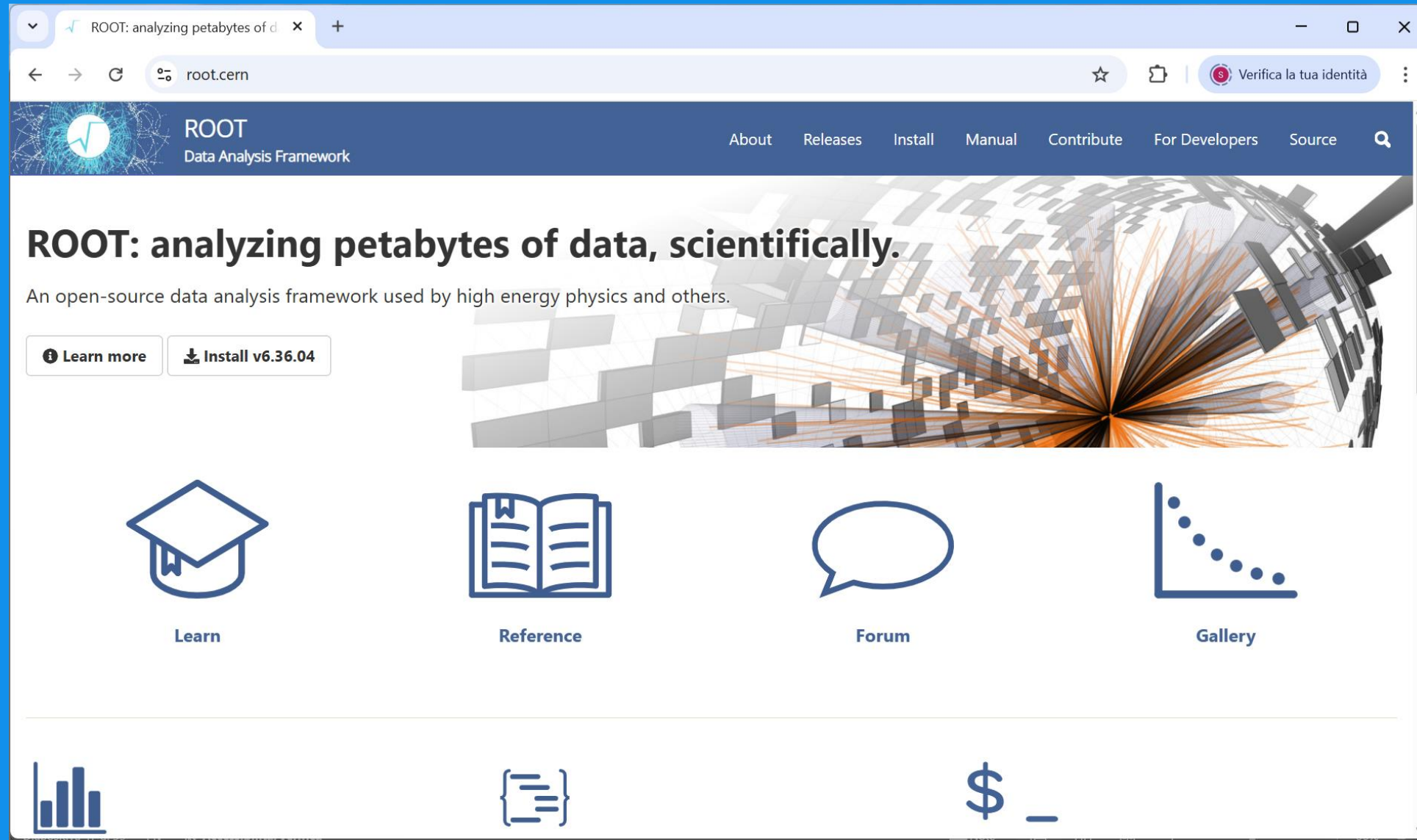
```
G4 kernel has come to Quit state.  
Deleting G4Run (id:1)  
UserDetectorConstruction deleted 0x5ee3f71aff80  
UserPhysicsList deleted 0x5ee3f71b0c10  
UserActionInitialization deleted 0x5ee3f736c000  
UserWorkerInitialization deleted 0  
UserWorkerThreadInitialization deleted 0x5ee3f8696a90  
UserRunAction deleted.  
UserPrimaryGenerator deleted.  
RunManager is deleting RunManagerKernel.  
G4SDManager deleted.  
EventManager deleted.  
Units table cleared.  
TransportationManager deleted.  
Total navigation history collections cleaned: 3  
G4RNGHelper object is deleted.  
===== Deleting memory pools =====  
Pool ID '20G4NavigationLevelRep', size : 0.00288 MB  
Pool ID '24G4ReferenceCountedHandleIvE', size : 0.000961 MB  
Pool ID '17G4DynamicParticle', size : 0.000961 MB  
Pool ID '15G4CountedObjectIvE', size : 0.000961 MB  
Number of memory pools allocated: 4; of which, static: 0  
Dynamic pools deleted: 4 / Total memory freed: 0.0058 MB  
===== G4Allocator objects are deleted.  
UImanager deleted.  
StateManager deleted.  
RunManagerKernel is deleted. Good bye :)  
RunManager is deleted.  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_b
```

Prova Esemplio B1

[illegible]

ROOT

<https://root.cern/>

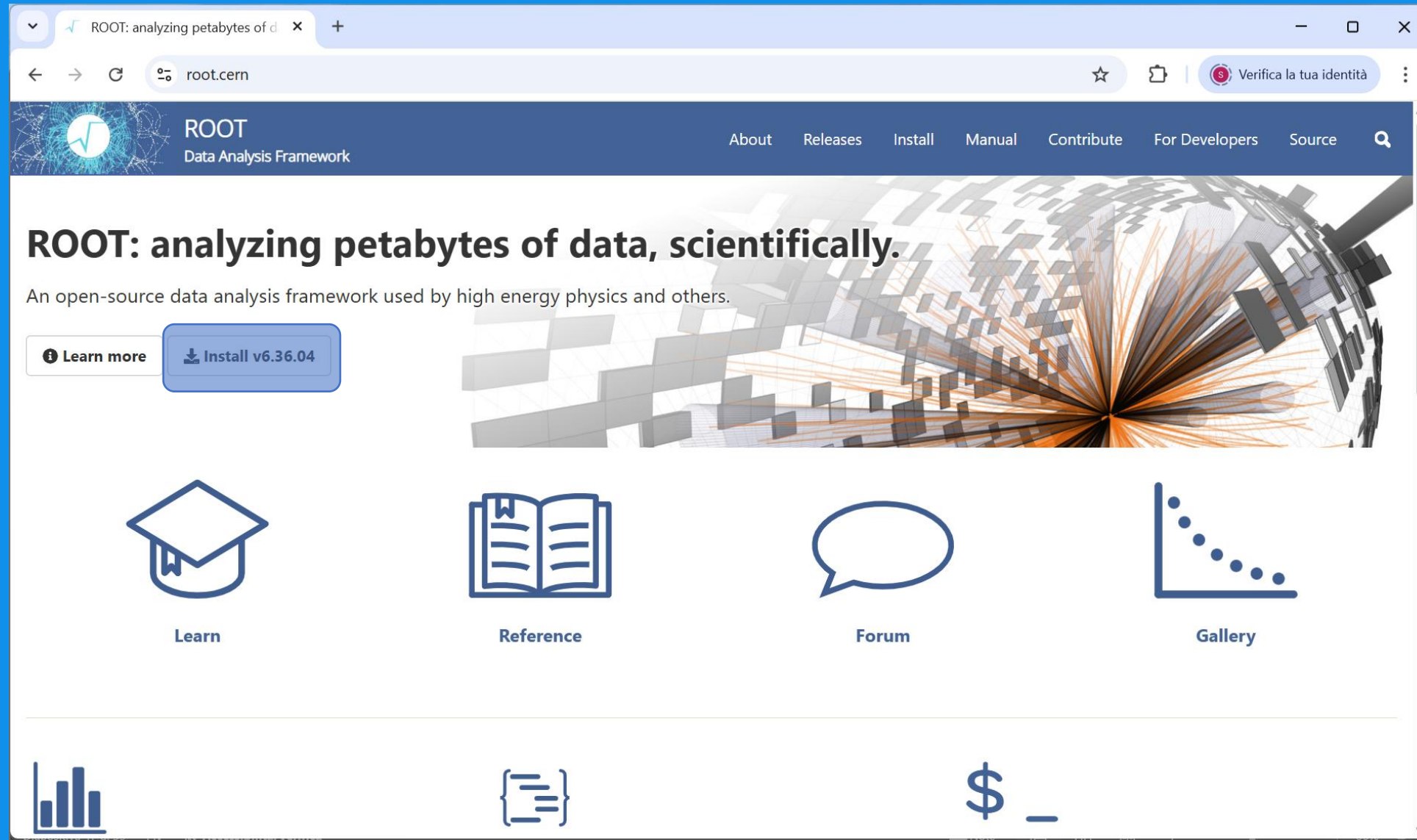


ROOT - Prerequisiti

[illegible]

ROOT

<https://root.cern/>



ROOT

<https://root.cern/>

Installing ROOT

ROOT is available on Linux, Mac, and Windows. To install it, we highly recommend (ordered by priority) either:

1. Use a package manager
2. Download a pre-compiled binary

Further ways to install ROOT on your computer are listed in the table of content on the right. Which one is best for you depends on your operating system and usage requirements.

In all cases, make sure to always use the most recent ROOT release possible to get the latest bug fixes, features and quick user support. The latest stable ROOT release is 6.32.04 (about ROOT versioning scheme).

Download a pre-compiled binary distribution

We distribute pre-compiled ROOT for several major Linux distributions as well as MacOS and (as a beta) Windows. The steps to install a pre-compiled binary are simple:

1. Install all [required dependencies](#) with the system package manager
2. [Download the release](#) for the desired platform and ROOT version
3. Unpack the archive
4. Add the ROOT libraries and executables to your environment by sourcing the appropriate `thisroot.*` script. These setup scripts can be found in the ROOT binary release, in the `bin` directory.



Click poi selezionare la versione «Latest Stable»

ROOT - Download

Release 6.36.04 - 2025-08-25

Release Notes

The release notes for this release can be found [here](#).

Binary distributions

Instead of manually downloading this binary, please explore first whether your [package manager](#) already provides this version. This way, you will automatically keep up-to-date with the latest stable versions with no manual maintenance on your side.

Platform	Files	Size
Almalinux 8.10	root_v6.36.04.Linux-almalinux8.10-x86_64-gcc8.5.tar.gz	276M
Almalinux 9.6	root_v6.36.04.Linux-almalinux9.6-x86_64-gcc11.5.tar.gz	296M
Almalinux 10.0	root_v6.36.04.Linux-almalinux10.0-x86_64-gcc14.2.tar.gz	305M
Debian 12	root_v6.36.04.Linux-debian12-x86_64-gcc12.2.tar.gz	283M
Fedora 41	root_v6.36.04.Linux-fedora41-x86_64-gcc14.3.tar.gz	285M
Fedora 42	root_v6.36.04.Linux-fedora42-x86_64-gcc15.2.tar.gz	304M
Ubuntu 22.04	root_v6.36.04.Linux-ubuntu22.04-x86_64-gcc11.4.tar.gz	286M
Ubuntu 24.04	root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz	282M
Ubuntu 25.04	root_v6.36.04.Linux-ubuntu25.04-x86_64-gcc14.2.tar.gz	285M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.pkg	408M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.tar.gz	268M



Click tasto destro

Useful links

All releases

Release 6.36.04 - 2025-08-25

Release Notes

The release notes for this release can be found [here](#).

Binary distributions

Instead of manually downloading this binary, please explore first whether your [package manager](#) already provides this version. This way, you will automatically keep up-to-date with the latest stable versions with no manual maintenance on your side.

On this page

Release 6.36.04 - 2025-08-25
Release Notes
Binary distributions
Source distribution
Installations in CVMFS
Git
Windows
Important installation notes

Platform	Files
Almalinux 8.10	root_v6.36.04.Linux-almalinux8.10-x86_64-gcc13.3.tar.gz
Almalinux 9.6	root_v6.36.04.Linux-almalinux9.6-x86_64-gcc13.3.tar.gz
Almalinux 10.0	root_v6.36.04.Linux-almalinux10.0-x86_64-gcc13.3.tar.gz
Debian 12	root_v6.36.04.Linux-debian12-x86_64-gcc13.3.tar.gz
Fedora 41	root_v6.36.04.Linux-fedora41-x86_64-gcc13.3.tar.gz
Fedora 42	root_v6.36.04.Linux-fedora42-x86_64-gcc13.3.tar.gz
Ubuntu 22.04	root_v6.36.04.Linux-ubuntu22.04-x86_64-gcc13.3.tar.gz
Ubuntu 24.04	root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz
Ubuntu 25.04	root_v6.36.04.Linux-ubuntu25.04-x86_64-gcc14.2.tar.gz
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang15.pkg

Apri link in un'altra scheda

Apri link in un'altra finestra

Apri link in finestra di navigazione in incognito

Salva link con nome...

Copia indirizzo link

Ispeziona

ROOT – Installazione

```
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
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sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ mkdir ROOT  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ cd ROOT/  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$ wget https://root.cern/download/root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz  
--2025-11-01 11:20:35-- https://root.cern/download/root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz  
Resolving root.cern (root.cern)... 137.138.6.72, 2001:1458:201:8b::100:23b  
Connecting to root.cern (root.cern)|137.138.6.72|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 296152492 (282M) [application/x-gzip]  
Saving to: 'root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz'  
  
root_v6.36.04 57%[=====]> ] 163.52M 1.80MB/s eta 66s
```


ROOT – Installazione

```
sg@LAPTOP-4KU4SPH9: ~/RC × + ▾  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ mkdir ROOT  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ cd ROOT/  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$ wget https://root.cern/download/root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz  
--2025-11-01 11:20:35-- https://root.cern/download/root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz  
Resolving root.cern (root.cern)... 137.138.6.72, 2001:1458:201:8b::100:23b  
Connecting to root.cern (root.cern)|137.138.6.72|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 296152492 (282M) [application/x-gzip]  
Saving to: 'root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz'  
  
root_v6.36.04.Linux-ubuntu24. 100%[=====>] 282.43M 1.14MB/s in 2m 47s  
  
2025-11-01 11:23:25 (1.69 MB/s) - 'root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz' saved [296152492/296152492]  
  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$ ls  
root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$ tar -xvzf root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz
```

ROOT – Installazione

```
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$ ls  
root  root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz  
sg@LAPTOP-4KU4SPH9: ~/ROOT$  
sg@LAPTOP-4KU4SPH9: ~/ROOT$ ls -l root  
total 144  
-rw-r--r--  1 sg sg   989 Aug 25 11:08 LICENSE  
drwxr-xr-x  3 sg sg  4096 Aug 25 11:43 README  
drwxr-xr-x  2 sg sg  4096 Aug 25 11:43 bin  
drwxr-xr-x  3 sg sg  4096 Aug 25 11:43 cmake  
drwxr-xr-x  2 sg sg  4096 Aug 25 11:43 config  
drwxr-xr-x  8 sg sg  4096 Aug 25 11:43 etc  
drwxr-xr-x  2 sg sg  4096 Aug 25 11:43 fonts  
drwxr-xr-x  3 sg sg  4096 Aug 25 11:43 geom  
drwxr-xr-x  2 sg sg 20480 Aug 25 11:43 icons  
drwxr-xr-x 16 sg sg 61440 Aug 25 11:43 include  
drwxr-xr-x  7 sg sg  4096 Aug 25 11:43 js  
drwxr-xr-x 10 sg sg 12288 Aug 25 11:43 lib  
drwxr-xr-x  2 sg sg  4096 Aug 25 11:43 macros  
drwxr-xr-x  3 sg sg  4096 Aug 25 11:43 man  
drwxr-xr-x 14 sg sg  4096 Aug 25 11:43 tutorials  
drwxr-xr-x 10 sg sg  4096 Aug 25 11:43 ui5  
sg@LAPTOP-4KU4SPH9: ~/ROOT$
```

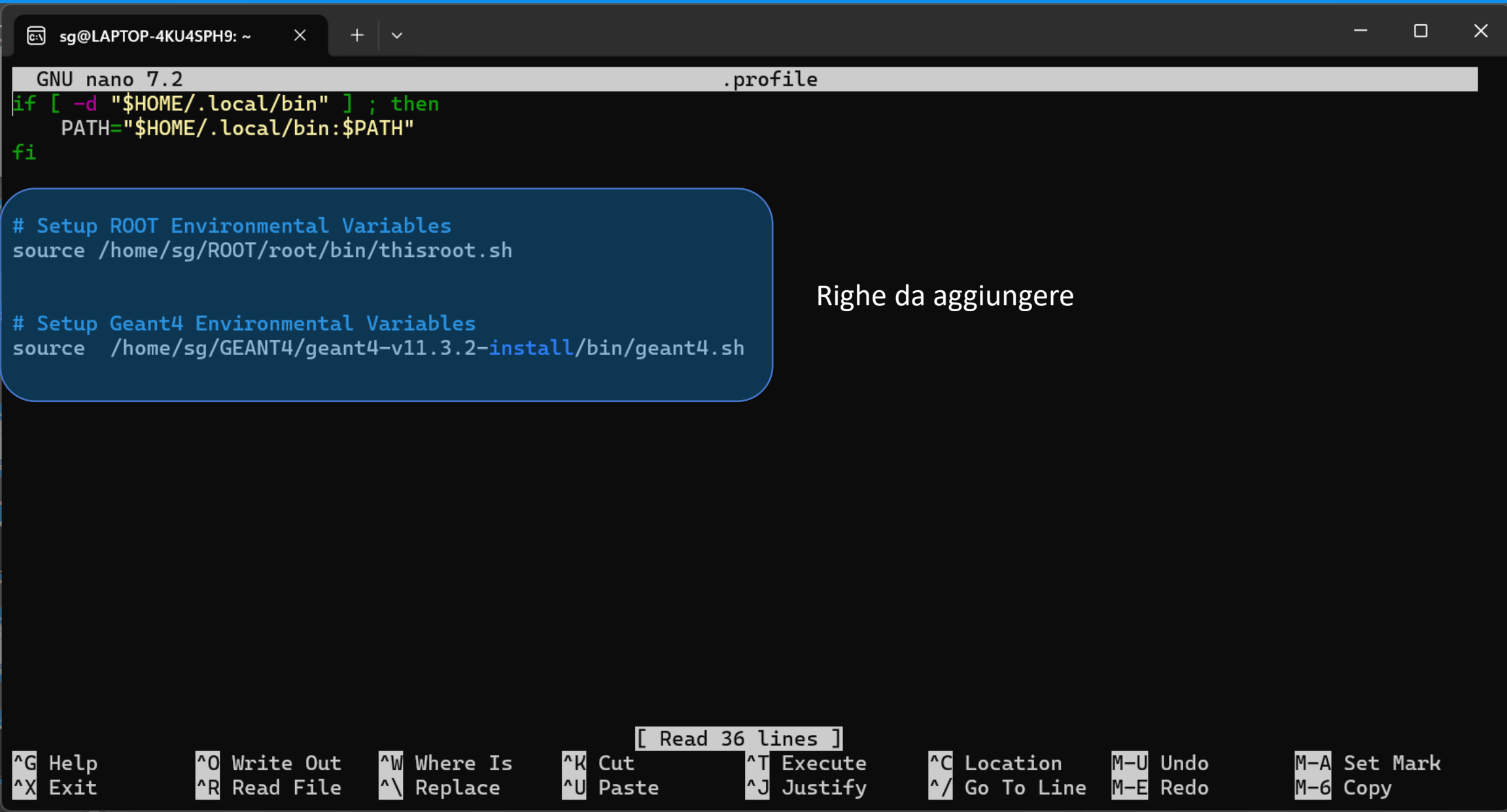
ROOT – controllo installazione

```
sg@LAPTOP-4KU4SPH9: ~/R/ × + ▾  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$ source root/bin/thisroot.sh  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$ root -l  
root [0]  
root [0]  
root [0] .q  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$ python3  
Python 3.12.3 (main, Aug 14 2025, 17:47:21) [GCC 13.3.0] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>>  
>>>  
>>> import ROOT  
>>>  
>>>  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$
```

ROOT – attivazione automatica

[illegible]

ROOT e GEANT 4 – attivazione automatica



```
GNU nano 7.2 .profile
if [ -d "$HOME/.local/bin" ] ; then
  PATH="$HOME/.local/bin:$PATH"
fi

# Setup ROOT Environmental Variables
source /home/sg/ROOT/root/bin/thisroot.sh

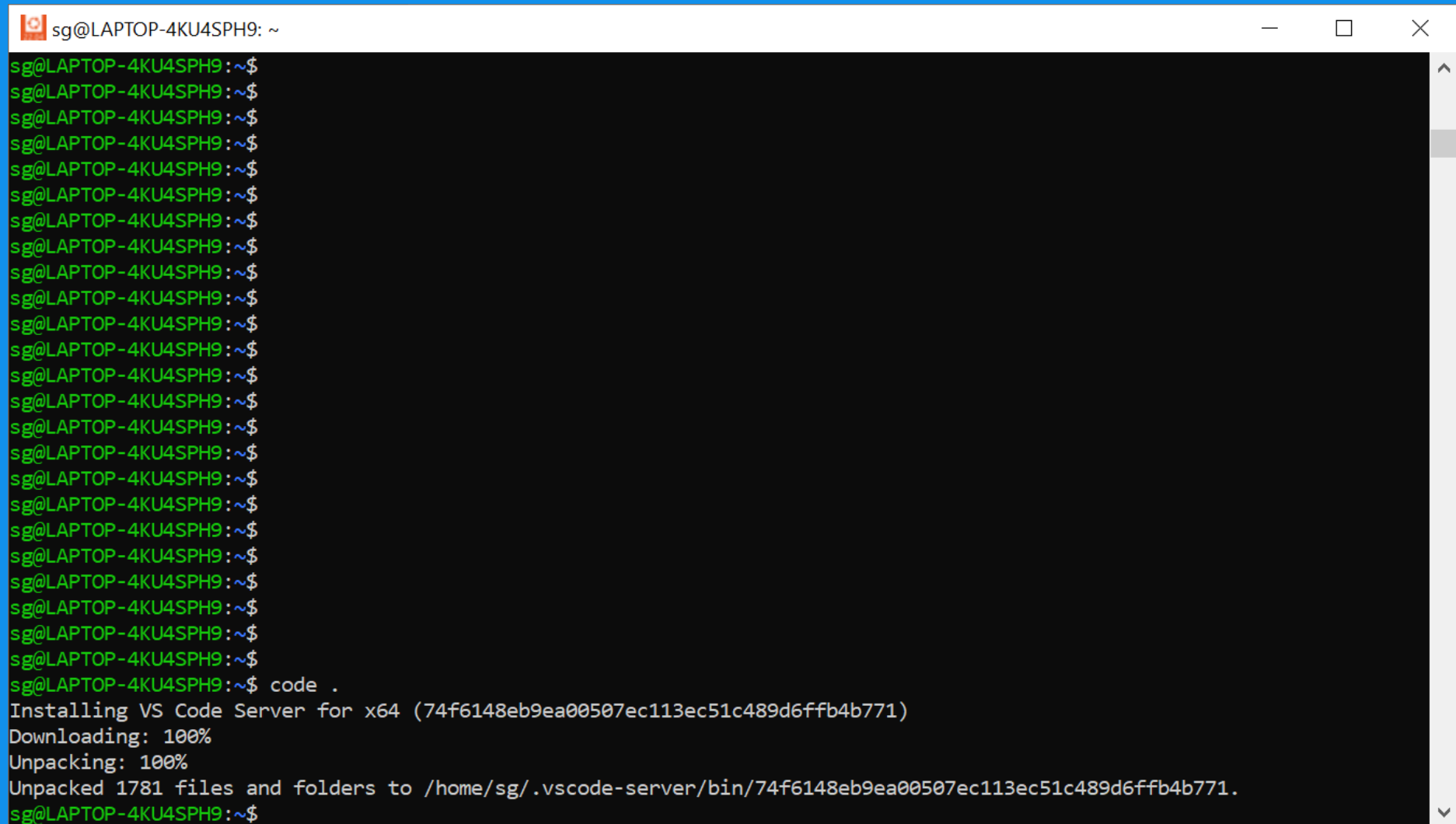
# Setup Geant4 Environmental Variables
source /home/sg/GEANT4/geant4-v11.3.2-install/bin/geant4.sh
```

Righe da aggiungere

[Read 36 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy

WSL – Visual Studio Code (da Windows)



The screenshot shows a terminal window titled "sg@LAPTOP-4KU4SPH9: ~". The terminal displays a series of 20 identical prompts "sg@LAPTOP-4KU4SPH9: ~\$" followed by the command "code .". After the command is executed, the terminal shows the following output:

```
Installing VS Code Server for x64 (74f6148eb9ea00507ec113ec51c489d6ffb4b771)
Downloading: 100%
Unpacking: 100%
Unpacked 1781 files and folders to /home/sg/.vscode-server/bin/74f6148eb9ea00507ec113ec51c489d6ffb4b771.
sg@LAPTOP-4KU4SPH9: ~$
```